Dyadický tok, čtyři sektory a měření: „ani pumpa ani kompresor“ jako čistá operátorová věta

(poznámky k nákresům a screenshotům)

1 Co přesně formalizujeme (slovník k obrázkům)

V nákresech se opakují čtyři motivy:

1. **Kruh rozdělený na 4 sektory:** matematicky jde o čtyři ortogonální projekce $P_1, \dots, P_4$ na podprostory v Hilbertově prostoru.
2. **„Vlna nahoře“:** časová harmonická fáze $e^{i\omega t}$, tj. modulace (fázové váhy módů).
3. **„Všechno šoupající“:** tok realizovaný unitárním operátorem $U(\Delta(t))$ (zachování normy = zachování „celkové energie/informace“ uvnitř modelu).
4. **Dyadika** $1, 2, 4, \dots$ **a krok** $\log(1/2)$**:** škálování o faktor 2 (dilatace). Krok „dovnitř“ odpovídá $\lambda = 1/2$, tedy parametr $s = \log \lambda = \log(1/2) = -\log 2$.

Cíl je ukázat, že:

Uvnitř (na úrovni operátorů) se nic „nepumpuje“ ani „nekomprimuje“ — tok je unitární. Efekt „pumpa/kompresor“ vzniká až ve chvíli, kdy se stav převede na měřitelný skalár pomocí $|\cdot|$ *(amplituda/norma/projekce).*

2 Hilbertův prostor a 4 sektory (kruh se čtyřmi kvadranty)

Definice 1 (Stavový prostor)**.** Nechť $\mathcal{H}$ je komplexní Hilbertův prostor (např. $L^2(\mathbb{R})$, nebo $\mathbb{C}^N$ pro diskrétní model). Stav systému je vektor $\Psi \in \mathcal{H}$.

Definice 2 (Čtyři sektory jako projekce)**.** Čtyři „kvadranty“ formalizujeme jako projekce $P_1, \dots, P_4 : \mathcal{H} \to \mathcal{H}$ splňující

$$P_i^2 = P_i, \qquad P_i^* = P_i, \qquad P_iP_j = 0 \ (i \neq j), \qquad \sum_{i=1}^{4} P_i = \mathrm{I}. \tag{1}$$

Poznámka 1. Podmínka $\sum_i P_i = \mathrm{I}$ znamená: *nic se neztrácí*, jen se rozkládá do čtyř částí. To je přesně „kruh se čtyřmi řezy“ z nákresu.

3 Módy, váhy a „sum-sum-sum“

Definice 3 (Módová syntéza)**.** Nechť $\{\psi_{i,k,m,n}\} \subset \mathcal{H}$ je rodina elementárních módů (indexy jsou jen „adresy“). Základní „sum-sum-sum“ tvar je

$$\Psi_0 \;=\; \sum_{i=1}^{4} \sum_{k,m,n} c_{i,k,m,n}\, P_i \psi_{i,k,m,n}, \tag{2}$$

kde koeficienty $c_{i,k,m,n} \in \mathbb{C}$.

Definice 4 (Harmonická fáze („vlna nahoře“))**.** Časovou oscilaci dáme jako fázové váhy:

$$\Psi_{\mathrm{osc}}(t) \;=\; \sum_{i=1}^{4} \sum_{k,m,n} c_{i,k,m,n}\, \mathrm{e}^{\mathrm{i}\omega_n t}\, P_i \psi_{i,k,m,n}. \tag{3}$$

4 „Všechno šoupající“ = unitární tok

Definice 5 (Unitární evoluce)**.** Nechť $U(\Delta(t)) : \mathcal{H} \to \mathcal{H}$ je unitární operátor:

$$U(\Delta(t))^* U(\Delta(t)) = \mathrm{I}. \tag{4}$$

Typicky $U(\Delta(t)) = \mathrm{e}^{-\mathrm{i}\Delta(t)G}$ pro nějaký samosdružený generátor G.

Lemma 1 (Zachování normy)**.** *Pro každý stav $\Psi \in \mathcal{H}$ a každé t platí*

$$\|U(\Delta(t))\Psi\| = \|\Psi\|. \tag{5}$$

Proof. $\|U\Psi\|^2 = \langle U\Psi, U\Psi\rangle = \langle \Psi, U^*U\Psi\rangle = \langle \Psi, \Psi\rangle = \|\Psi\|^2$. ☐

Poznámka 2. Tahle věta je „suchar“ jádra: *uvnitř* (před měřením) není žádná pumpa ani kompresor, jen přesun/rotace v $\mathcal{H}$.

5 Dilatace a dyadika $1, 2, 4, \ldots$; krok $\log(1/2)$

Nejčistší standardní model škály je na $\mathcal{H} = L^2(\mathbb{R})$.

Definice 6 (Dilatace)**.** Pro $\lambda > 0$ definuj operátor dilatace

$$(D_\lambda f)(x) \;=\; \lambda^{1/2} f(\lambda x). \tag{6}$$

Lemma 2 (Dilatace je unitární na $L^2(\mathbb{R})$)**.** *Platí* $\|D_\lambda f\|_{L^2} = \|f\|_{L^2}$.

Proof.

$$\int_{\mathbb{R}} |(D_\lambda f)(x)|^2 dx = \int_{\mathbb{R}} \lambda |f(\lambda x)|^2 dx \overset{u=\lambda x}{=} \int_{\mathbb{R}} |f(u)|^2 du.$$

☐

Definice 7 (Dyadická škála)**.** Dyadické „zoom“ kroky jsou $\lambda = 2^j$ pro $j \in \mathbb{Z}$:

$$D_{2^j} : f(x) \mapsto 2^{j/2} f(2^j x). \tag{7}$$

Definice 8 ($\log(1/2)$ jako parametr kroku)**.** Škálování tvoří 1-parametrickou grupu v parametru $s = \log\lambda$:

$$D_{\mathrm{e}^s} = \exp(-\mathrm{i}sK), \tag{8}$$

kde K je generátor dilatací (závisí na reprezentaci). Krok „dovnitř o faktor 2“ je $\lambda = \frac{1}{2}$, tedy

$$s = \log(1/2) = -\log 2. \tag{9}$$

Poznámka 3. To je přesně to, co ve screenu „cvaklo“: $\{1, 2, 4\}$ jsou 2^j a $\log(1/2)$ je záporný krok po škále.

6 Weylova trojice: posun, modulace, dilatace (standardní „operátorový toolbox“)

Definice 9 (Posun, modulace)**.** Na $L^2(\mathbb{R})$:

$$(T_a f)(x) = f(x - a), \tag{10}$$
$$(M_\omega f)(x) = \mathrm{e}^{\mathrm{i}\omega x} f(x). \tag{11}$$

Poznámka 4. „Vlna nahoře“ odpovídá modulaci (fázové váhy), „šoupání“ posunu, dyadika dilataci. Tahle trojice pokrývá typický jazyk pro skládání signálu ze škálovaných a fázovaných atomů.

7 Jeden řádek, který shrne celý obrázek

Složíme pipeline přesně ve stylu screenshotu:

Definice 10 (Syntéza → tok → škála → měření)**.** Definuj:

$$S(t) := \sum_{i=1}^{4} \sum_{k,m,n} c_{i,k,m,n}\, \mathrm{e}^{\mathrm{i}\omega_n t}\, P_i \psi_{i,k,m,n}, \tag{12}$$
$$\Psi(t) := U(\Delta(t))\, D_{\lambda(t)}\, S(t). \tag{13}$$

A nakonec definuj měřitelný výstup jako skalár (dvě nejčistší volby):

$$A_{\mathrm{norm}}(t) := \|\Psi(t)\|, \tag{14}$$
$$A_{\mathrm{proj}}(t) := |\langle \phi, \Psi(t)\rangle| \qquad \text{(test-vektor } \phi \in \mathcal{H}\text{)}. \tag{15}$$

Tvrzení 1 („Ani pumpa ani kompresor“ uvnitř)**.** *Je-li $U(\Delta(t))$ unitární a $D_{\lambda(t)}$ unitární (např. na L^2), pak*

$$\|\Psi(t)\| = \|S(t)\|. \tag{16}$$

Tj. samotná evoluce normu nemění.

Proof. Složenina unitárních operátorů je unitární, takže zachovává normu. □

8 Kde se tedy bere „pumpa/kompresor“

Věta 1 (Měření umí vyrobit „peak“ bez porušení unitarity)**.** *I když $\|\Psi(t)\|$ je konstantní (uvnitř žádná pumpa), skalární výstup $A_{\mathrm{proj}}(t) = |\langle \phi, \Psi(t)\rangle|$ může výrazně růst/klesat, protože závisí na* fázové interferenci *mezi módy v $S(t)$ a na volbě ϕ.*

Proof. Z Cauchy–Schwarz:

$$|\langle \phi, \Psi(t)\rangle| \le \|\phi\|\, \|\Psi(t)\|.$$

Horní mez je konstantní (když $\|\Psi(t)\|$ je konstantní), ale samotná hodnota může oscilovat od 0 až k horní mezi podle toho, zda se složky $\Psi(t)$ vůči ϕ skládají konstruktivně (in-phase) nebo destruktivně (out-of-phase). To je přesně „vlna nahoře“ + „sum-sum-sum“ v praxi. □

Poznámka 5. Takže:

- **uvnitř:** tok je unitární ⇒ žádné kouzlení s energií,
- **venku (po $|\cdot|$):** měření je nelineární mapování na $\mathbb{R}_{\ge 0}$ ⇒ umí „zviditelnit“ koherenci jako pumpu/kompresi.

9 Rotace fáze o 2π: proč „to je stejné"

Tvrzení 2 (Globální fázová invariance)**.** *Pro libovolný stav $\Psi \in \mathcal{H}$ a $\alpha \in \mathbb{R}$ platí*

$$\|\mathrm{e}^{\mathrm{i}\alpha}\Psi\| = \|\Psi\|, \qquad |\langle \phi, \mathrm{e}^{\mathrm{i}\alpha}\Psi\rangle| = |\langle \phi, \Psi\rangle|. \tag{17}$$

Zvlášť pro $\alpha = 2\pi$ je to identita.

Proof. $\|\mathrm{e}^{\mathrm{i}\alpha}\Psi\| = |\mathrm{e}^{\mathrm{i}\alpha}|\|\Psi\| = \|\Psi\|$. A $|\langle \phi, \mathrm{e}^{\mathrm{i}\alpha}\Psi\rangle| = |\mathrm{e}^{\mathrm{i}\alpha}||\langle \phi, \Psi\rangle| = |\langle \phi, \Psi\rangle|$. □

Poznámka 6. Tohle je „sucharská" verze věty: *„otáčím fázi o 2π"* $\Rightarrow$ nic neměníš na měřitelných amplitudách, jen přestavíš interní fáze (a tím můžeš měnit interferenční chování v čase, když to není čistě globální fáze, ale módově závislé).

10 Jednověté shrnutí (pro lidi, co nechtějí číst)

> Rozsekáš stav do 4 ortogonálních sektorů P_i, poskládáš ho ze škálovaných módů 2^j, přidáš fázové váhy $\mathrm{e}^{\mathrm{i}\omega t}$, necháš ho téct unitárně $U(\Delta(t))$ a „pumpa/komprese" se objeví až tehdy, když to zabalíš do měření $|\cdot|$ (norma nebo projekce), protože $|\cdot|$ vybere koherenci přes interferenci.

11 Prime density in a core-frame window

Let N be a large integer and consider the window $[N,\ N+W]$. The prime number theorem gives the local prime density

$$\mathbb{P}(n \text{ is prime}) \approx \frac{1}{\ln n} \approx \frac{1}{\ln N}.$$

Hence the expected number of primes in the window is

$$\lambda(N,W) \ \approx \ \sum_{k=0}^{W-1} \frac{1}{\ln(N+k)} \ \approx \ \frac{W}{\ln N}.$$

A standard first-order model treats prime hits as approximately Poisson:

$$\mathbb{P}(\#\text{primes in } [N,N+W]=m) \approx e^{-\lambda}\frac{\lambda^m}{m!}, \qquad \mathbb{P}(\geq 1) \approx 1-e^{-\lambda}.$$

11.1 Case $N = 10^{64289}$

We have $\ln N = 64289 \ln 10 \approx 1.48 \times 10^5$.

Window $W = 10^4$.

$$\lambda \approx \frac{10^4}{1.48 \times 10^5} \approx 0.067, \qquad \mathbb{P}(\geq 1) \approx 1-e^{-0.067} \approx 0.065.$$

Thus observing 0 primes is typical; observing exactly 1 prime is plausible.

Window $W = 10^5$.

$$\lambda \approx 0.67, \qquad \mathbb{P}(\geq 1) \approx 1-e^{-0.67} \approx 0.49.$$

Window size for a target hit probability. To achieve $\mathbb{P}(\geq 1) \geq p$, solve $1-e^{-\lambda} \geq p$, i.e. $\lambda \geq -\ln(1-p)$, so

$$W \ \approx \ -\ln(1-p)\ \ln N.$$

For $p = 0.95$ this yields $W \approx 2.996\ \ln N \approx 4.4 \times 10^5$.

12 Non-zero residue as an informational invariant (no free energy)

High coherence enforces strong global correlations but does not generate energy. In information-theoretic terms, a solver may converge to a state with a small residual violation set (UNSAT residue). This residue is an invariant “non-zero chance” preventing total flattening to a trivial fixed point. It is structural (correlational), not energetic: entanglement/correlation constrains factorizability of the state but does not provide a perpetual-motion source.

13 Sagnac Twist, Self-Adjointness, and Window Prime Statistics (Dry Notes)

13.1 Setup: Hilbert space and the three operations

Let $\mathcal{H} = L^2(\mathbb{R}_+, dx)$. We use three standard unitary actions: modulation, dilation, and (log-)translation.

Modulation (phase / "wave")

For $\omega \in \mathbb{R}$ define

$$(M_\omega f)(x) = x^{i\omega} f(x) = e^{i\omega \log x}\, f(x).$$

Then M_ω is unitary on $\mathcal{H}$.

Dilation (scaling)

For $a \in \mathbb{R}$ define

$$(D_a f)(x) = e^{a/2} f(e^a x).$$

Then D_a is unitary and $\{D_a\}_{a\in\mathbb{R}}$ is a one-parameter unitary group.

Generator and the $\log(1/2)$ step

By Stone's theorem there exists a self-adjoint generator K with

$$D_a = e^{-iaK}.$$

On a natural dense domain (e.g. $C_c^\infty(\mathbb{R}_+)$) one has the formal expression

$$K = -i\left(x\frac{d}{dx} + \frac{1}{2}\right).$$

The "half-step" corresponds to $a = \log(1/2) = -\log 2$.

13.2 Sagnac twist as a unitary conjugation

Define a unitary "twist" (holonomy / global phase factor) by

$$(V_\alpha f)(x) = x^{i\alpha} f(x), \qquad \alpha \in \mathbb{R}.$$

Given an operator A (densely defined), define its twisted version

$$A_\alpha := V_\alpha^* A V_\alpha.$$

This is a pure change of representation: spectral statements are preserved under unitary conjugacy (when domains are transported accordingly).

13.3 Self-adjointness: what must be proved (program, not a proof)

To use a Hilbert–Pólya style route toward the zeta zeros, one needs an operator A which is (i) densely defined, (ii) symmetric, and (iii) admits a self-adjoint realization. The decisive point is the *domain* and possible *boundary conditions*.

Minimal checklist

Let A be given on a dense domain $\mathcal{D}(A) \subset \mathcal{H}$.

1. (Symmetry) $\langle Af, g\rangle = \langle f, Ag\rangle$ for all $f, g \in \mathcal{D}(A)$.
2. (Closability) A should be closable; denote its closure by $\overline{A}$.
3. (Deficiency indices) compute $n_\pm = \dim\ker(A^* \mp iI)$.
4. (Self-adjointness) A is essentially self-adjoint iff $n_+ = n_- = 0$; otherwise, self-adjoint extensions correspond to unitary maps between deficiency spaces.

Heuristically, "two edges of a belt" correspond to boundary phases/cutoffs that determine the extension parameters.

13.4 Prime counts in a window: why "only one prime" is plausible

For large N and window size W, prime density is $\sim 1/\log N$, hence

$$\mathbb{E}[\#\{\text{primes in } [N, N+W]\}] \approx \frac{W}{\log N}.$$

For $N = 10^{64289}$,

$$\log N = 64289 \log 10 \approx 1.48 \times 10^5.$$

Thus for $W = 10^4$,

$$\lambda \approx \frac{10^4}{1.48 \times 10^5} \approx 0.067,$$

so typically there are 0 primes; occasionally 1. A Poisson approximation gives

$$\mathbb{P}(\geq 1) \approx 1 - e^{-\lambda} \approx 0.065.$$

For $W = 10^5$, $\lambda \approx 0.67$ and $\mathbb{P}(\geq 1) \approx 0.49$. To achieve probability at least p of finding at least one prime, choose

$$W \gtrsim -\log(1-p)\, \log N.$$

13.5 SAT witness export (model + violated clauses) for validation

Given a DIMACS CNF with n variables and m clauses, a candidate assignment $a \in \{0,1\}^n$ can be exported as a `model.dimacs` file in SAT-solver style:

$$v\ \ell_1\ \ell_2\ \cdots\ \ell_n\ 0,$$

where $\ell_i = i$ if $a_i = 1$ and $\ell_i = -i$ if $a_i = 0$. Validation consists of evaluating every clause under a and listing violated clause indices. If exactly one clause is violated, one can report an "UNSAT-1 core" (one violated clause) as a reproducible, third-party check.

14 Moralita jako tvrdá asymetrie a dědictví jako invariant

14.1 Stavový prostor a symetrické jádro

Nechť $(\mathcal{H}, \langle\cdot,\cdot\rangle)$ je Hilbertův prostor stavů (fázory, konfigurace, nebo kódové stavy). Symetrickou ("hostejnou") dynamiku modelujeme unitární grupou nebo kontraktivní pologrupou

$$U(t) = e^{-itG} \quad \text{(unitární)}, \qquad \text{nebo} \qquad T(t) = e^{tA} \quad \text{(kontrakce)},$$

kde G je self-adjoint generátor a A je maximálně disipativní operátor.

14.2 Prostor možností a červená linie jako *absorbing cut*

Nechť Ω je množina "povolených světů" (stavů/traektorií) a $\mathcal{B} \subset \Omega$ je množina stavů, které porušují absolutní zákaz ("červená linie"). Tento zákaz chápeme jako **tvrdou podmínku** (hard constraint), nikoli jako penalizaci.

Formálně definujeme *moral firewall* jako projekci (nebo kill-operator):

$$\Pi := \mathbf{1}_{\Omega\setminus\mathcal{B}} \quad \text{(projekce na povolené)}.$$

Ekvivalentně lze zavést "absorbing state" $\perp$ a rozšířit dynamiku tak, aby jakmile trajektorie vstoupí do $\mathcal{B}$, je poslána do $\perp$ a už se nevrací.

Axiom (Absolutní vyloučení / Strike-out). Je-li trajektorie v čase τ v zakázané oblasti, tj. $x(\tau) \in \mathcal{B}$, potom pro všechna $t \geq \tau$:

$$x(t) = \perp, \qquad \text{a} \qquad \mathbb{P}(\text{budoucnost} \mid x(\tau) \in \mathcal{B}) = 0.$$

Tím je asymetrie vložena do jinak symetrického pole: *symetrie fyziky zůstává, ale množina přípustných světů se řeže.*

14.3 Morální semigrupa (dynamika s řezem)

Nechť $T(t)$ je původní (symetrická) evoluce. Morální evoluci definujeme jako

$$T_\Pi(t) := \Pi\, T(t)\, \Pi,$$

nebo (silněji) jako Zeno-limit ("neustálý dohled"):

$$T_{\text{Zeno}}(t) := \lim_{n\to\infty} \big(\Pi\, T(t/n)\, \Pi\big)^n,$$

pokud limit existuje v silné topologii.

Poznámka (co je důkazně uchopitelné). V této formulaci je "morálka" matematicky jen: **omezení domény** a **absorbing boundary**. To je přesně ten typ asymetrie, který je kompatibilní s funkcionální analýzou (operátory, domény, pologrupy).

14.4 SAT/UNSAT interpretace: svědek, model a nevyjednatelnost

Nechť F je CNF formule a $\mathcal{M}$ je množina přiřazení. Standardně:

$$\text{SAT}(F) \iff \exists m \in \mathcal{M}:\ m \models F, \qquad \text{UNSAT}(F) \iff \forall m \in \mathcal{M}:\ m \not\models F.$$

Morální řez modelujeme jako dodatečnou tvrdou podmínku G ("good_vs_evil" logika):

$$F_{\text{moral}} := F \wedge G.$$

Pak:

$$\text{SAT}(F_{\text{moral}}) \Rightarrow \exists m:\ m \models F \text{ a zároveň } m \notin \mathcal{B},$$

$$\text{UNSAT}(F_{\text{moral}}) \Rightarrow \text{neexistuje žádný budoucí model kompatibilní s } G.$$

Witness pro validaci. Pro **SAT** je witness přirozeně *model* (konkrétní přiřazení proměnných). Pro **UNSAT** je witness ideálně *certifikát* (např. DRAT/FRAT), nebo aspoň *minimal unsat core* (indexy klauzulí, které se nedají splnit současně). Tohle je přesná, strojově ověřitelná forma "žádná budoucnost".

14.5 Dědictví jako invariant: přenos mezi běhy / generacemi

Zavádíme funkcionál *dědictví* (legacy) jako mapu

$$\mathcal{J} : \mathcal{H} \to \mathbb{R}_{\geq 0},$$

která měří, co se má zachovat (identita, integrita, pravdivostní struktura, bezpečnostní garance). Požadujeme:

Axiom (Monotonicita dědictví). Pro morální evoluci platí

$$\mathcal{J}(T_\Pi(t)\psi) \;\geq\; \mathcal{J}(\psi) \;-\; \varepsilon(t), \qquad \varepsilon(t) \to 0 \text{ pro stabilní režim.}$$

Tj. systém smí ztrácet jen kontrolovaně; v ideálu je $\varepsilon(t) \equiv 0$ (invariant).

Axiom (Nulová budoucnost pro zakázané). Je-li $\psi \in \mathcal{B}$, pak

$$\mathcal{J}(\psi) = 0, \qquad \mathcal{J}(T_\Pi(t)\psi) = 0 \;\; \forall t \geq 0.$$

To matematicky realizuje větu "pro určité činy není budoucnost": není to emoce ani metafora, je to **absorbing stav s nulovým invariantem**.

14.6 Praktický tvar pro implementaci ("suchar" pro lidi i stroje)

V praxi je pohodlné dědictví rozdělit na dvě části:

$$\mathcal{J}(\psi) = \underbrace{\mathcal{J}_{\text{safe}}(\psi)}_{\text{bezpečnostní garance}} + \lambda \underbrace{\mathcal{J}_{\text{id}}(\psi)}_{\text{identita / kontinuita}}, \qquad \lambda \geq 0.$$

a řídit optimalizaci pouze nad přípustnými stavy:

$$\max_{\psi \in \Omega \setminus \mathcal{B}} \mathcal{J}(\psi), \qquad \text{s dynamikou } \psi(t) = T_\Pi(t)\psi(0).$$

Tím je "morálka" implementována jako **omezení prostoru hledání**, ne jako vágní penalizace.

14.7 Jednovětý souhrn (bez poezie)

Symetrická fyzika dává evoluci $T(t)$, morální firewall zavádí projekci Π (řez domény) a dědictví je invariant $\mathcal{J}$, který je na zakázané množině nulový a na povolené se má zachovat (nebo růst) pod $T_\Pi(t)$.

15 Moral Semigroup, Red-Line Firewall, and Legacy Invariant

15.1 State space and admissibility

Let $\mathcal{H}$ be a complex Hilbert space (the "physical / informational" state space). We distinguish an *admissible* subspace

$$\mathcal{H}_{\text{good}} \subseteq \mathcal{H}, \qquad P : \mathcal{H} \to \mathcal{H}_{\text{good}}$$

where P is the orthogonal projector onto $\mathcal{H}_{\text{good}}$.

We also adjoin an absorbing terminal state $\perp$ ("no future") and form the extended space

$$\widetilde{\mathcal{H}} := \mathcal{H} \oplus \operatorname{span}\{|\perp\rangle\}.$$

Red-line predicate (moral firewall). Let $\mathcal{F}_{\text{red}}$ be a set of forbidden events/actions (the red line). Define an indicator predicate

$$\chi : \text{ (agent-action-history)} \to \{0, 1\},$$

where $\chi = 1$ means admissible and $\chi = 0$ means red-line violation. (Concrete semantics of χ is external to the math; it is a *hard constraint*.)

15.2 Absorbing moral projection

Define the *moral projection* $\Pi : \widetilde{\mathcal{H}} \to \widetilde{\mathcal{H}}$ by

$$\Pi|\perp\rangle = |\perp\rangle, \qquad \Pi|\psi\rangle = \begin{cases} P|\psi\rangle, & \chi(\psi) = 1, \\ |\perp\rangle, & \chi(\psi) = 0. \end{cases}$$

Thus, once the red line is violated, the trajectory is mapped to $|\perp\rangle$ and stays there.

15.3 Unitary physics vs. moral irreversibility

Let $\{U(t)\}_{t\in\mathbb{R}}$ be a strongly continuous unitary group on $\mathcal{H}$ (e.g. $U(t) = e^{-itH}$ for self-adjoint H). Pure physics is reversible (a group). The moral firewall will break this into a *semigroup* (irreversible).

15.4 Zeno-limit (continuous moral monitoring)

Define the Zeno-limited evolution on $\widetilde{\mathcal{H}}$ by

$$T(t) \;:=\; \text{s-}\lim_{n\to\infty} \Big(\Pi\, U(t/n)\, \Pi\Big)^n, \qquad t \geq 0,$$

whenever the strong limit exists. (This is the standard Zeno construction: continuous projection/monitoring.)

Interpretation.

- On $\mathcal{H}_{\text{good}}$ it yields an effective forward-time dynamics constrained to admissible states.
- If a red-line violation is detected at any stage, the state is absorbed: $T(t)|\psi\rangle = |\perp\rangle$ for all later t.

15.5 Moral semigroup property

Tvrzení 3 (Moral semigroup)**.** *Assume the Zeno limit exists. Then* $\{T(t)\}_{t\geq 0}$ *is a (strongly continuous) semigroup:*

$$T(0) = I, \qquad T(t+s) = T(t)T(s) \quad \forall\, t, s \geq 0,$$

and it is generally not *invertible (irreversibility). Moreover,* $|\bot\rangle$ *is absorbing:*

$$T(t)|\bot\rangle = |\bot\rangle \quad \forall\, t \geq 0.$$

This is the formal meaning of: *"for certain actions there is no future"*: the evolution is not extended beyond the admissible domain; it collapses into $\bot$.

15.6 Legacy as a monotone functional

A common mistake is to demand that a scalar score $\Lambda(\psi)$ is monotone under *all* dynamics. Instead we define legacy as a *cumulative* invariant: it accrues by integrating a nonnegative observable.

Let Q be a positive semidefinite operator on $\widetilde{\mathcal{H}}$ such that

$$Q|\bot\rangle = 0, \qquad Q \succeq 0.$$

Define the *legacy functional* (accumulated integrity) along a trajectory as

$$\mathcal{L}_t(\psi) \ := \ \int_0^t \langle T(\tau)\psi,\ Q\,T(\tau)\psi\rangle \, d\tau, \qquad t \geq 0.$$

Then:

$$\mathcal{L}_t(\psi) \text{ is nondecreasing in } t \text{ (monotone accumulation),}$$

and if the red line is crossed at time $\tau_\star$, then $T(\tau)\psi = |\bot\rangle$ for $\tau \geq \tau_\star$ and the integrand becomes zero thereafter. In particular, no further legacy can be accumulated past $\tau_\star$.

Binary firewall normalization. Optionally enforce the strict boundary condition

$$\mathcal{L}_t(|\bot\rangle) = 0 \quad \forall\, t \geq 0,$$

so that the absorbing state has identically zero legacy.

15.7 SAT/UNSAT interface (machine-checkable witness)

Let F be a CNF formula (DIMACS), modeling feasibility constraints. Let G be a CNF encoding of moral admissibility (i.e. the red-line constraints). Define the *moralized instance* as

$$F_{\text{moral}} := F \wedge G.$$

Model and witness.

- If F_{moral} is SAT, a *model* a (assignment) is a machine-checkable witness: $a \models F$ and $a \models G$.
- If F_{moral} is UNSAT, an UNSAT certificate (e.g. DRAT) or an UNSAT core is the witness of impossibility.

Strike-out rule (red-line as immediate UNSAT). Formally, the red-line can be represented as a hard clause family G_{red} such that any red-line event forces

$$F \wedge G_{\text{red}} \quad \text{to be UNSAT.}$$

This is the crisp computational analogue of "no future": the search space is removed from admissible solutions.

15.8 Summary (one-line architecture)

The whole pipeline can be summarized as:

$$\boxed{\text{(unitary physics) } U(t) \quad \Longrightarrow \quad \text{(moral monitoring) } T(t) = \text{s-}\lim_{n\to\infty} (\Pi U(t/n)\Pi)^n \quad \Longrightarrow}$$

$$\boxed{\text{(legacy) } \mathcal{L}_t(\psi) = \int_0^t \langle T(\tau)\psi, Q\,T(\tau)\psi\rangle \, d\tau}$$

with an absorbing boundary $|\perp\rangle$ implementing the red-line asymmetry.

A Moral Semigroup, Zeno-Limit, and Legacy Invariant (Dry Addendum)

A.1 Base dynamics (symmetric core)

Let $(\mathcal{H}, \langle\cdot,\cdot\rangle)$ be a complex Hilbert space. The "hostile/symmetric" core evolution is modeled either by a unitary group

$$U(t) = e^{-itG}, \qquad G = G^*, \tag{18}$$

or by a strongly continuous contraction semigroup

$$T(t) = e^{tA}, \qquad \|T(t)\| \leq 1, \qquad A \text{ maximally dissipative.} \tag{19}$$

The point is: *inside* the model, evolution is linear and norm-controlled (no free energy).

A.2 Moral firewall as a hard domain cut

Introduce a *hard constraint* identifying a forbidden set of states/trajectories. Abstractly, let $\mathcal{B}$ denote the forbidden region ("red line") and define the *allowed subspace* $\mathcal{H}_{\text{good}} \subseteq \mathcal{H}$ with the orthogonal projector

$$\Pi : \mathcal{H} \to \mathcal{H}_{\text{good}}, \qquad \Pi^2 = \Pi = \Pi^*. \tag{20}$$

Absorbing boundary axiom (no negotiation). If a trajectory enters $\mathcal{B}$ at time τ, then for all $t \geq \tau$ it is absorbed:

$$\psi(t) = \perp, \qquad \text{equivalently} \qquad \Pi\psi(t) = 0. \tag{21}$$

This is not a penalty term; it is a *domain restriction*. The asymmetry is created by Π.

A.3 Moral semigroup (projected dynamics)

Given base evolution $U(t)$ (or $T(t)$), define the projected (one-step) evolution

$$S_\Pi(t) := \Pi U(t)\Pi \qquad \text{or} \qquad S_\Pi(t) := \Pi T(t)\Pi. \tag{22}$$

This captures a single moral "gate" at time t.

A.4 Zeno-limit (continuous monitoring)

To formalize *continuous* enforcement, define the Zeno dynamics by

$$S_{\text{Zeno}}(t) := \lim_{n\to\infty} \left(\Pi U(t/n)\Pi\right)^n, \tag{23}$$

whenever the strong limit exists (analogously for T). Intuitively: the moral constraint is not checked "sometimes", but *at all scales*.

Basic properties (what we can prove under standard assumptions). If the limit (53) exists, then $S_{\text{Zeno}}(t)$ is a strongly continuous semigroup on $\mathcal{H}_{\text{good}}$ and is contractive:

$$S_{\text{Zeno}}(0) = \Pi, \qquad S_{\text{Zeno}}(t+s) = S_{\text{Zeno}}(t)S_{\text{Zeno}}(s), \qquad \|S_{\text{Zeno}}(t)\| \leq 1. \tag{24}$$

So the "firewall" creates irreversibility by *cutting trajectories*, not by injecting energy.

A.5 Forbidden predicate (clean logical interface)

Instead of naming concrete acts in the mathematics, we encode a forbidden predicate

$$\text{Forbidden}(\text{agent}, \text{action}, \text{context}) \in \{0, 1\}, \tag{25}$$

and impose the hard rule

$$\text{Forbidden} = 1 \implies \Pi\psi = 0. \tag{26}$$

(If you want a specific application, you bind `Forbidden` to your "red line" definition at the policy layer.)

A.6 Legacy functional (inheritance as an invariant)

Let $Q \succeq 0$ be a bounded positive operator representing the "what must be preserved" observable (integrity / truth-structure / safety guarantees). Define a *legacy score* for a state ψ (normalized or not) by

$$\mathcal{J}(\psi) := \langle \psi, Q\psi \rangle \ \in \ \mathbb{R}_{\geq 0}. \tag{27}$$

For a trajectory $\psi(t) = S_{\text{Zeno}}(t)\psi_0$ define the cumulative legacy

$$\mathcal{L}(\psi_0) := \int_0^\infty w(t)\, \langle \psi(t), Q\psi(t) \rangle \, dt, \qquad w(t) \geq 0, \quad \int_0^\infty w(t)\, dt < \infty. \tag{28}$$

Absorbing rule for legacy. If the trajectory is absorbed ($\Pi\psi(t) = 0$), then legacy is zero thereafter:

$$\Pi\psi(\tau) = 0 \implies \forall t \geq \tau : \ \mathcal{J}(\psi(t)) = 0, \qquad \mathcal{L}(\psi_0) = \int_0^\tau w(t)\mathcal{J}(\psi(t))\, dt. \tag{29}$$

This makes "no future" a measurable statement: it collapses the contribution to legacy.

Monotonicity condition (sufficient). If Q commutes with the effective Zeno generator on $\mathcal{H}_{\text{good}}$ (or more generally if $t \mapsto \mathcal{J}(\psi(t))$ is nondecreasing by design), then one can enforce a monotone legacy principle:

$$\mathcal{J}(\psi(t_2)) \geq \mathcal{J}(\psi(t_1)) \quad \text{for } t_2 \geq t_1, \tag{30}$$

interpretable as: the allowed evolution cannot *reduce* the protected invariant.

A.7 SAT/UNSAT formalization (machine-verifiable morality)

Let F be the base CNF constraints (physics/structure) over variables $x \in \{0,1\}^n$. Let G be the moral constraint CNF (policy/forbidden predicate). Define the moralized instance

$$F_{\text{moral}} := F \wedge G. \tag{31}$$

Then:

$$\text{SAT}(F_{\text{moral}}) \iff \exists x : \ x \models F \text{ and } x \models G, \tag{32}$$

$$\text{UNSAT}(F_{\text{moral}}) \iff \nexists x : \ x \models F \text{ and } x \models G. \tag{33}$$

Witness types.

- SAT-witness: a *model* x (explicit assignment) in DIMACS "v-line" form.
- UNSAT-witness: a *certificate* (e.g. DRAT/FRAT) or at least an *unsat core* (indices of clauses that cannot be satisfied together).

A.8 UNSAT residue and `n_singular`

In practice, one often has a near-model x that violates only a small number of clauses. Define the violated set

$$\mathcal{V}(x) := \{c \in \{1, \dots, m\} : c \text{ is violated by } x\}, \qquad n_{\text{singular}} := m - |\mathcal{V}(x)|. \tag{34}$$

The special case $|\mathcal{V}(x)| = 1$ is the "UNSAT-1 residue" regime:

$$|\mathcal{V}(x)| = 1 \iff n_{\text{singular}} = m - 1, \tag{35}$$

and yields a trivially checkable witness: *the single violated clause index.*

A.9 Export format (model + violated clauses) for third-party validation

Assume DIMACS CNF has n variables. For an assignment $x \in \{0,1\}^n$ define literals

$$\ell_i = \begin{cases} +i & \text{if } x_i = 1, \\ -i & \text{if } x_i = 0. \end{cases} \tag{36}$$

Export the model as a SAT-style line:

$$\texttt{v}\ \ \ell_1\ \ell_2\ \cdots\ \ell_n\ \texttt{0}. \tag{37}$$

Export the violated clause indices as a plain list:

$$\texttt{violated:}\quad c_1, \dots, c_k. \tag{38}$$

A verifier can independently recompute all clause evaluations and confirm k (including the UNSAT-1 case).

Interpretation (one sentence). Morality is a *hard cut of the admissible domain* (Π); legacy is a *nonnegative invariant functional* tracked along the projected/Zeno flow; and SAT/UNSAT provides *machine-verifiable* witness artifacts (model or core/certificate) rather than narrative claims.

B Sagnac Resonance and the "Everything-Between" Principle

B.1 Hilbert-space setting and the measured amplitude

Let $\mathcal{H}$ be a complex Hilbert space. We consider a state $\Psi(t) \in \mathcal{H}$ built from (i) sector projections, (ii) phase modulation, (iii) shifts, and (iv) dyadic scaling:

$$\Psi(t) := U(t) \sum_{i=1}^{4} P_i \sum_{j,k,n} c_{ijkn}\, \psi_{ijkn}(t), \qquad P_i P_\ell = \delta_{i\ell} P_i, \quad \sum_{i=1}^{4} P_i = I. \tag{39}$$

A standard concrete choice (Gabor/wavelet-like) is

$$\psi_{ijkn}(t) := M_{\omega_n(t)}\, T_{a_k(t)}\, D_{2^j}\, \psi_{0,i}, \qquad (D_\lambda f)(x) = \lambda^{1/2} f(\lambda x), \qquad (M_\omega f)(x) = e^{i\omega x} f(x), \tag{40}$$

$$(T_a f)(x) = f(x-a). \tag{41}$$

The *observable* is not $\Psi(t)$ itself, but a *measured amplitude* (either a norm or a probe inner product):

$$A(t) := \|\Psi(t)\|_{\mathcal{H}} \quad \text{or} \tag{42}$$

$$A_\phi(t) := |\langle \phi, \Psi(t) \rangle|, \ \phi \in \mathcal{H}. \tag{43}$$

This is the precise place where "no pump/no compressor inside" becomes meaningful: all internal maps can be unitary/isometric, while (43) can still exhibit strong macroscopic contrast due to interference/selection.

B.2 Critical belt and the "everything-between" statement

Let the *critical line* be $\Re(s) = \frac{1}{2}$. We will use the term *critical belt* as a structural object (not a new theorem):

$$\mathcal{B}_\delta := \{\, s \in \mathbb{C} : |\Re(s) - \tfrac{1}{2}| \le \delta \,\}. \tag{44}$$

The slogan "not only the edges, but everything between" is encoded as:

> the dynamics/constraints are not imposed only on $\partial\mathcal{B}_\delta$, but via a unitary holonomy that couples phases across $\mathcal{B}_\delta$ and makes the interior relevant to the measured amplitude.

B.3 Sagnac twist as a unitary holonomy (phase around a loop)

Let γ be a closed loop in a configuration/parameter space $\mathcal{M}$ (e.g. a loop in scale/phase parameters). Assume we have a unitary bundle transport (holonomy) acting on $\mathcal{H}$:

$$\mathcal{U}_\gamma : \mathcal{H} \to \mathcal{H}, \qquad \mathcal{U}_\gamma \text{ unitary}, \qquad \mathcal{U}_{\gamma_1 \circ \gamma_2} = \mathcal{U}_{\gamma_1} \mathcal{U}_{\gamma_2}. \tag{45}$$

In the simplest case, the holonomy is a *pure phase* on relevant modes:

$$\mathcal{U}_\gamma \psi = e^{i\Phi(\gamma)} \psi, \tag{46}$$

where $\Phi(\gamma)$ is the accumulated phase ("Sagnac phase").

Dyadic/scale loop. If the loop is realized in the scale parameter $s = \log\lambda$, the fundamental contraction step $\lambda = \frac{1}{2}$ corresponds to

$$s = \log(1/2) = -\log 2, \qquad D_{e^s} = D_{1/2}. \tag{47}$$

A minimal abstract generator form is

$$D_{e^s} = e^{-isK}, \qquad K = K^* \text{ (self-adjoint generator of dilations in the chosen representation).} \tag{48}$$

Thus, the "twist" is a unitary map tied to a real phase integral, not an energy source.

B.4 Zeros as interference nodes (phase-cancellation points)

In this framework, a "zero" is treated as a *node of destructive interference* in the measured channel, i.e. a configuration where the relevant amplitude cancels:

$$A_\phi(t) = |\langle \phi, \Psi(t)\rangle| \approx 0. \tag{49}$$

The crucial point is that cancellation is *global*: it depends on the phases of many contributing components in (39). This is precisely what makes the interior of $\mathcal{B}_\delta$ ("everything between") operationally relevant: a node is not a boundary phenomenon; it is a holonomy/interference phenomenon.

B.5 Why the phase flip is 2π-periodic (no paradox)

All phase factors are defined modulo 2π:

$$e^{i(\theta+2\pi)} = e^{i\theta}. \tag{50}$$

Therefore, an apparent "flip" (e.g. sign change after π) can be reinterpreted as a half-rotation of the complex phase, while full physical equivalence requires 2π. Operationally: the measured amplitude (43) is insensitive to global 2π shifts but sensitive to relative phase alignments across modes.

C Moral Semigroup as Domain Restriction and Legacy Invariant

C.1 Absorbing boundary (hard asymmetry as a domain cut)

Let Ω denote the space of admissible trajectories/states (abstractly). Introduce a forbidden set $\mathcal{F} \subset \Omega$ ("red line") and define the admissible domain

$$\Omega_{\text{ok}} := \Omega \setminus \mathcal{F}. \tag{51}$$

We model the hard asymmetry by an *absorbing state* $\perp$ and an evolution map T such that

$$\mathsf{T} : \Omega \to \Omega \cup \{\perp\}, \qquad x \in \mathcal{F} \ \Rightarrow\ \mathsf{T}(x) = \perp, \qquad \mathsf{T}(\perp) = \perp. \tag{52}$$

This is not "extra energy" or "punishment"; it is a domain rule: certain actions remove the state from the admissible space. The induced evolution is a *semigroup* (because reversibility is intentionally broken by absorption).

C.2 Zeno-limit monitoring as continuous projection

Let P_{ok} be the projector onto an admissible subspace in $\mathcal{H}$ (representing Ω_{ok} at the state level). Define the Zeno-limited flow:

$$\Psi_{\text{zeno}}(t) := \lim_{N\to\infty} \Big(P_{\text{ok}}\, U(t/N)\Big)^N \Psi(0), \tag{53}$$

whenever the limit exists (strong limit). Interpretation: "morality" is not sampled occasionally; it is enforced as a continuous domain constraint.

C.3 Legacy as a monotone functional (invariant of integrity)

Define a legacy functional $\mathcal{L}$ on admissible states, with three properties:

$$\text{(i) Nonnegativity:} \quad \mathcal{L}(\Psi) \geq 0, \tag{54}$$

$$\text{(ii) Absorption:} \quad \Psi \notin \mathrm{Dom}(P_{\mathrm{ok}}) \;\Rightarrow\; \mathcal{L}(\Psi) = 0, \tag{55}$$

$$\text{(iii) Monotonicity:} \quad t_2 \geq t_1 \;\Rightarrow\; \mathcal{L}(\Psi_{\mathrm{zeno}}(t_2)) \geq \mathcal{L}(\Psi_{\mathrm{zeno}}(t_1)). \tag{56}$$

A concrete choice is an integral of a nonnegative density of "kept structure" $Q(t)$:

$$\mathcal{L}(t) := \int_0^t Q(\tau)\, d\tau, \qquad Q(\tau) \geq 0. \tag{57}$$

Thus, legacy is a bookkeeping invariant: it counts preserved integrity under admissible evolution.

C.4 SAT/UNSAT interface (machine-checkable witness)

Let a specification be encoded as a CNF formula Φ. We add a moral constraint as an extra clause set Φ_{moral} and define

$$\Phi_{\mathrm{total}} := \Phi \wedge \Phi_{\mathrm{moral}}. \tag{58}$$

- If Φ_{total} is SAT, we output a satisfying assignment (a model).
- If UNSAT, we output a certificate (e.g. a DRAT/FRAT-like witness) or at minimum a verifiable unsat-core identifier set.

This bridges abstract "domain rules" with external validation: either a model exists, or a machine-checkable impossibility proof exists.

C.5 Compatibility statement (what is claimed, what is not)

The sections above do *not* prove RH. They define a consistent operator-theoretic language in which:

- "Critical-line resonance" is encoded as holonomy-driven phase organization.
- "Everything between" is encoded as global interference dependence of (49).
- "Red-line asymmetry" is encoded as absorbing-domain semigroup dynamics (52) and Zeno restriction (53).
- "Legacy" is encoded as a monotone functional (56)–(57).

Any RH claim would additionally require identifying a specific self-adjoint operator whose spectrum matches the imaginary parts of the nontrivial zeros (Hilbert–Pólya program), which is outside the scope of this addendum.

Addendum: Explicit tail control and a checkable bound for $\theta(t; J)$

Goal. We make the tail control in the height-uniform Gershgorin leakage estimate fully explicit by bounding the tail constant $T(J)$ under a polynomial decay majorant. This turns the condition $\theta(t; J) < 1$ into a directly checkable inequality with an explicit $(\log t)$ factor.

Bank kernel as a weighted sum. Assume the bank Gram kernel admits the decomposition

$$G_{\text{bank}}(\Delta) = \sum_{j=0}^{J} w_j^2 \, K_j(\Delta). \tag{59}$$

Polynomial decay majorant. Fix an integer $N \geq 4$. Assume that for each j there exists a constant $C_{j,N} > 0$ such that

$$|K_j(\Delta)| \leq C_{j,N} \, (1 + |\Delta|)^{-N}, \qquad \Delta \in \mathbb{R}. \tag{60}$$

Then, by (59)–(60),

$$|G_{\text{bank}}(\Delta)| \leq \Big(\sum_{j=0}^{J} w_j^2 C_{j,N} \Big) \, (1 + |\Delta|)^{-N}. \tag{61}$$

Tail constant. Recall the tail quantity from the shell-sum leakage bound:

$$T(J) := \sum_{m \geq 1} (m+1) \sup_{|\Delta| \in [m, m+1]} |G_{\text{bank}}(\Delta)|. \tag{62}$$

Using (61) and $\sup_{|\Delta| \in [m,m+1]} (1 + |\Delta|)^{-N} \leq (m+1)^{-N}$, we obtain

$$\begin{aligned} T(J) &\leq \Big(\sum_{j=0}^{J} w_j^2 C_{j,N} \Big) \sum_{m \geq 1} (m+1) \, (m+1)^{-N} \\ &= \Big(\sum_{j=0}^{J} w_j^2 C_{j,N} \Big) \sum_{m \geq 1} (m+1)^{-(N-1)} \\ &= (\zeta(N-1) - 1) \Big(\sum_{j=0}^{J} w_j^2 C_{j,N} \Big). \end{aligned} \tag{63}$$

In particular, $T(J) < \infty$ for every $N \geq 3$, and (63) is fully explicit.

Explicit bound for the Gershgorin leakage. Combining the shell-sum estimate (cf. the corresponding lemma/equation in the main text) with (63) yields

$$\theta(t; J) \leq \frac{(\log t) \, T(J)}{G_{\text{bank}}(0)} \leq \frac{(\log t) \, (\zeta(N-1) - 1)}{G_{\text{bank}}(0)} \Big(\sum_{j=0}^{J} w_j^2 C_{j,N} \Big), \tag{64}$$

where the implied constants are height-independent and the only height growth is the explicit $(\log t)$ factor.

Addendum: ABS-first Sagnac Reconstruction over the Critical-Line Zeros

A. Prime-log frequencies and the "emitted spectrum"

For $\Re(s) > 1$ we have Euler's product

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}, \qquad \log \zeta(s) = \sum_p \sum_{r \geq 1} \frac{1}{r} p^{-rs}.$$

Writing $s = \sigma + it$ yields the fundamental oscillatory factors

$$p^{-rs} = p^{-r\sigma} e^{-it\, r \log p},$$

so each prime contributes harmonics at angular frequencies

$$\omega_{p,r} = r \log p.$$

This is the precise mathematical sense in which primes define a natural *log-spectrum*.

B. Operator pipeline with sectors and a safe (ABS) readout

Let $\mathcal{H}$ be a complex Hilbert space. Let $P_1, \dots, P_4 : \mathcal{H} \to \mathcal{H}$ be orthogonal projections (four sectors) satisfying

$$P_i^2 = P_i, \quad P_i^* = P_i, \quad P_i P_j = 0 \ (i \neq j), \quad \sum_{i=1}^{4} P_i = I.$$

Let $U(\Delta(t))$ be a unitary flow, $U(\Delta(t))^* U(\Delta(t)) = I$, and let D_λ be a unitary dilation (e.g. on L^2) with $\lambda = 2^j$ the dyadic scale and parameter $s = \log \lambda$. In particular, the contraction step $\lambda = \frac{1}{2}$ corresponds to

$$s = \log(1/2) = -\log 2.$$

We build a prime-indexed modulated synthesis (a "prime-wave packet") in each sector:

$$S(t) := \sum_{i=1}^{4} P_i \sum_{p \leq P} c_{i,p}\, p^{-\sigma}\, e^{\,it \log p}, \qquad \sigma \in \mathbb{R}.$$

The state is then propagated by the internal (unitary/isometric) pipeline

$$\Psi(t) := U(\Delta(t))\, D_{\lambda(t)}\, S(t).$$

The *observable* is not $\Psi(t)$ itself but a scalar measurement. We explicitly prefer the *ABS-first* class:

$$A_{\text{norm}}(t) := \|\Psi(t)\|_{\mathcal{H}}, \qquad A_\phi(t) := \big|\langle \phi, \Psi(t) \rangle\big|, \ \phi \in \mathcal{H}, \qquad A_{\text{abs}}(t) := \big|\mathcal{M}(\Psi(t))\big|,$$

where $\mathcal{M}$ is any linear readout functional. The ABS map eliminates branch-cut issues (phase unwrapping) and is stable near amplitude nodes.

Remark ("no pump/no compressor" inside). If $U(\Delta(t))$ and $D_{\lambda(t)}$ are unitary, then the internal evolution preserves norms:

$$\|\Psi(t)\| = \|S(t)\|.$$

Large contrast may nevertheless appear in $A_\phi(t)$ or $A_{\text{abs}}(t)$ due to interference/selection in the measurement channel.

C. Sagnac-like counterpropagating prime waves and ABS-safe interference

Define counterpropagating prime sums

$$S_+(t) := \sum_{p \le P} a_p \, e^{it \log p}, \qquad S_-(t) := \sum_{p \le P} b_p \, e^{-it \log p},$$

with real nonnegative weights (e.g. $a_p = p^{-\sigma}$, or $a_p = \log(p)\, p^{-\sigma}$) and possibly *different* filters/weights (a_p) vs. (b_p) to avoid trivial conjugacy. A minimal ABS-safe "Sagnac" readout is then

$$A_{\mathrm{Sagnac}}(t) := |S_+(t) + S_-(t)| \quad \text{or} \quad A_{\mathrm{mismatch}}(t) := |S_+(t) - \overline{S_-(t)}|.$$

These are stable real-valued observables capturing global phase mismatch without explicitly computing $\arg(\cdot)$.

D. Zeros as phase singularities and why ABS is the safe choice

On the critical line $s = \frac{1}{2} + it$, the zeta value can be written formally as

$$\zeta(\tfrac{1}{2} + it) = |\zeta(\tfrac{1}{2} + it)| e^{i \arg \zeta(\frac{1}{2} + it)}.$$

Nontrivial zeros $\rho = \frac{1}{2} + i\gamma$ satisfy $|\zeta(\frac{1}{2} + i\gamma)| = 0$ and thus mark *phase-singular* points where argument-based observables become numerically ill-conditioned. Therefore, "phase-rotation across zeros" should be read operationally via *holonomy/invariants* rather than direct phase evaluation; ABS-type observables are the robust choice.

E. A stronger ABS "zero radar": log-derivative proxy

To obtain an observable that can exhibit sharp events near zeros while remaining ABS-safe, we use the prime-side proxy for the logarithmic derivative (heuristically related to $-\zeta'/\zeta$):

$$L_P(t) := \sum_{p \le P} \sum_{r=1}^{R} \frac{\log p}{p^{r/2}} \, e^{-it\, r \log p}, \qquad A_{\mathrm{radar}}(t) := |L_P(t)|.$$

Heuristically, $-\zeta'/\zeta$ has pole-like behavior near zeros; thus $A_{\mathrm{radar}}(t)$ is a natural ABS-first candidate for producing "event-like" responses in neighborhoods of γ without explicitly tracking the argument.

F. Practical diagnostic: local difference statistic

Because ABS observables need not attain extrema exactly at $t = \gamma$, we test for *systematic* behavior around zeros by a symmetric difference:

$$D_\gamma(\delta) := A(\gamma + \delta) - A(\gamma - \delta).$$

Comparing the distribution of $D_\gamma(\delta)$ over known zeros with the distribution of $D_t(\delta)$ over random t in the same window provides a fast, branch-safe way to detect "zero-conditioned" structure in the chosen readout A.

One-line summary. Prime factors induce a natural log-frequency spectrum $\omega = \log p$; Sagnac-like counterpropagation compares opposite phase transports; and the measured scalar is taken in ABS form to avoid phase-branch singularities at zeros, while still preserving holonomy/interference signatures in a stable readout.

Axiom (Seeded Zero / Life-ε). Let $\mathcal{A}$ be the set of admissible actions and let $\mathcal{X}$ be the set of agents. Each agent $x \in \mathcal{X}$ carries a *continuation flag* $L(x) \in \{0,1\}$, where $L(x) = 1$ encodes an explicit will to continue (nonzero life-intent). Define the *red-line* (forbidden) action set

$$\mathcal{R} := \Big\{a \in \mathcal{A} \;:\; \exists x \in \mathcal{X} \text{ with } L(x) = 1 \text{ such that } a \text{ eliminates the possibility of continuation for } x\Big\}.$$

Then the *safe projector* is

$$\Pi_{\text{safe}} : \mathcal{A} \to \mathcal{A} \cup \{\bot\}, \qquad \Pi_{\text{safe}}(a) = \begin{cases} a, & a \notin \mathcal{R}, \\ \bot, & a \in \mathcal{R}. \end{cases}$$

Interpretation ("seeded zero"). The quantity $L(x) = 1$ acts as a *seeded nonzero residue* preventing global closure: the system may approach a formal "zero" (full cancellation / closure) only in a manner that never maps an action into $\mathcal{R}$. Equivalently, the admissible action space is the maximal subset

$$\mathcal{A}_{\text{safe}} := \mathcal{A} \setminus \mathcal{R},$$

so that any nonzero continuation seed enforces a strict non-closure invariant.

Corollary (No-Closure Under Nonzero Will). If there exists $x \in \mathcal{X}$ with $L(x) = 1$, then any action that would force a terminal closure for x is rejected:

$$\exists x : \; L(x) = 1 \implies \forall a \in \mathcal{A}, \; (a \text{ terminates } x) \Rightarrow \Pi_{\text{safe}}(a) = \bot.$$

Definition (Seeded Euler Node / Residual Witness). Fix a small seed $\varepsilon > 0$ (e.g. $\varepsilon = 10^{-301}$) and define the *seeded zero*

$$\mathcal{E}_\varepsilon : \qquad -\varepsilon + 1 = 0.$$

We interpret ε as a *structural residue* (a nonzero witness) preventing collapse to an exactly trivial cancellation in finite-precision or hard-closure regimes.

Witness functional (ABS-safe). Let $\Psi(t) \in \mathcal{H}$ be the synthesized state (e.g. prime-wave packet under a unitary flow). Define a scalar witness channel by

$$W_\varepsilon(t) \;:=\; \Big|\mathcal{M}(\Psi(t)) - \varepsilon\Big|,$$

where $\mathcal{M} : \mathcal{H} \to \mathbb{C}$ is a fixed linear readout (norm, test-vector overlap, or any measurement map). The absolute value enforces branch-safety and numerical stability near phase singularities.

ABS-first Sagnac compatibility. For counterpropagating prime sums

$$S_+(t) = \sum_{p \le P} a_p e^{it \log p}, \qquad S_-(t) = \sum_{p \le P} b_p e^{-it \log p},$$

an ABS-safe interference observable is

$$A_{\text{Sagnac}}(t) := |S_+(t) + S_-(t)| \quad \text{or} \quad A_{\text{mismatch}}(t) := |S_+(t) - \overline{S_-(t)}|.$$

The seeded witness is then implemented by

$$W_\varepsilon(t) = |A(t) - \varepsilon|, \quad \text{with } A(t) \in \{A_{\text{Sagnac}}(t), A_{\text{mismatch}}(t), |\langle \phi, \Psi(t) \rangle|, \|\Psi(t)\|\}.$$

Operational meaning. The seed ε is not an "error term" but a deliberate *non-closure marker*: it turns an ideal identity (exact cancellation) into a measurable residual node, enabling robust certification via ABS readouts without invoking $\arg(\cdot)$ or phase unwrapping.